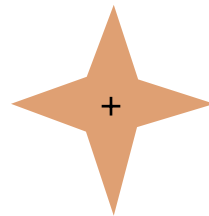
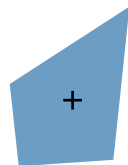
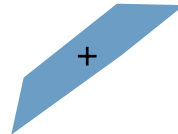
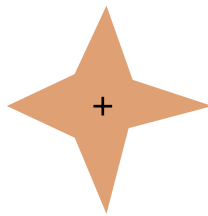
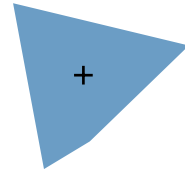
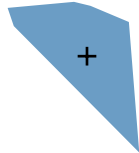
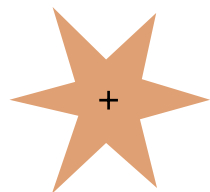
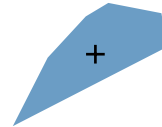
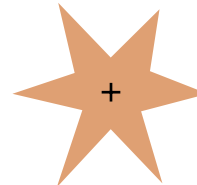
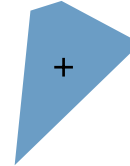
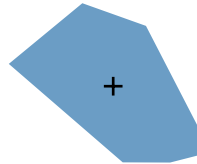
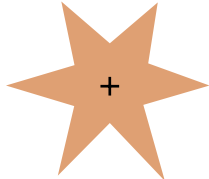
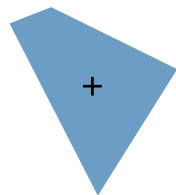
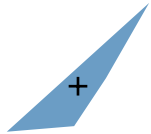
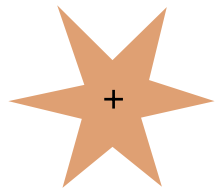
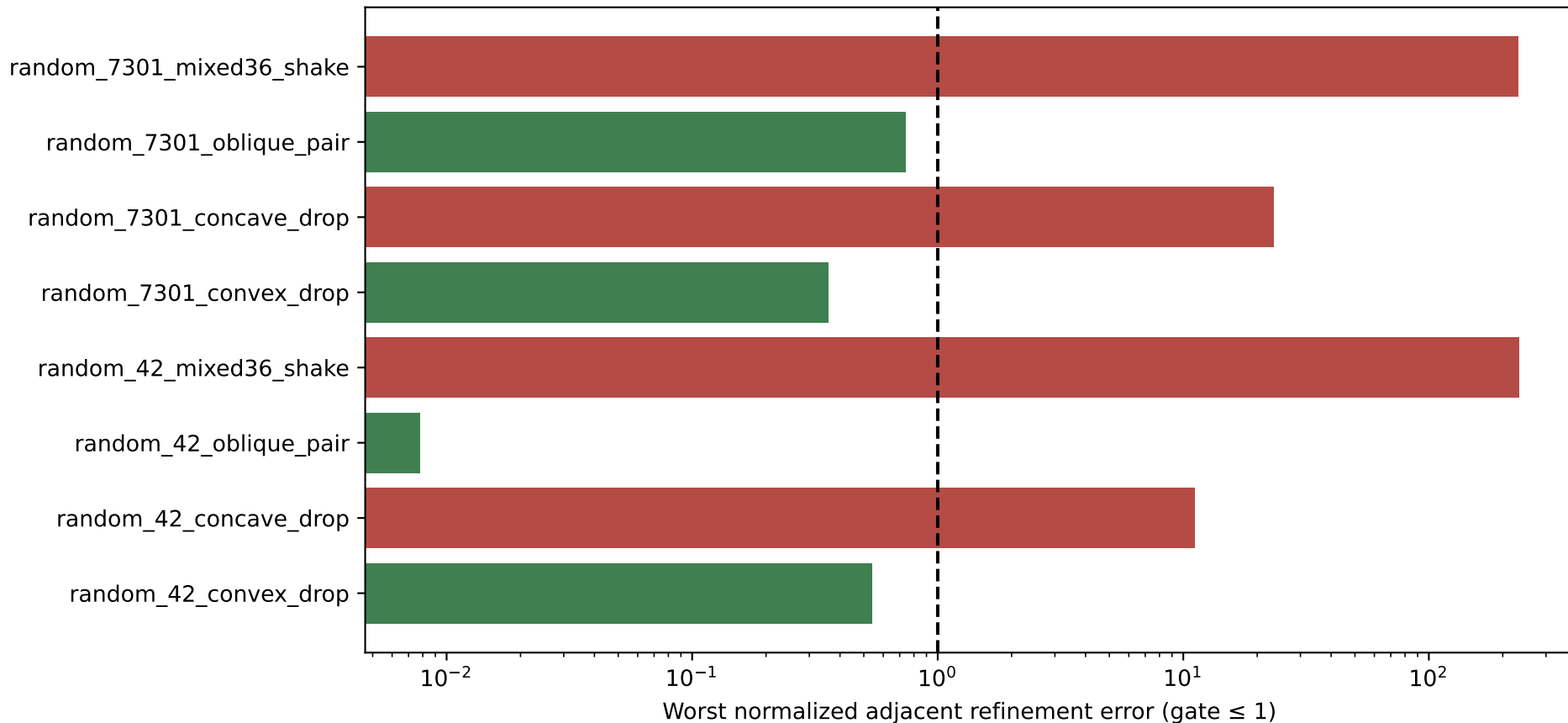


Actual seeded convex and concave outlines; + marks centre of mass



Reference qualification precedes any trajectory accuracy claim



Random-shape evidence — 5 October 2026

4/8 native references qualified; 12/40 candidates passed.

2 native geometry rejections retained.

24/24 frozen polygon contact solves accepted and independently audited.

Geometry: irregular convex hulls and concave triangle compounds; exact COM/inertia.

Full simulations: drops, off-centre pairs and 36 mixed shapes in shaking boxes.

Frozen contacts: real support vertices; normal/tangent/rotation coupling retained.

Limits: synthetic coefficients; no experimentally characterized materials.

Concave fixture skins/seams are a known modelling limitation.

Contact-chain verification is not full-world integration or arbitrary graph validation.

No random-shape dense-control speedup or universal engine claim.

Reproduce and audit: `research/run_random_shape_study.py`,
`research/audit_random_shape_study.py`. Full tables and rejection reasons: `report.md`.

Raw geometry, trajectories, contact states, timings and source hashes: `results/`.